BEFORE

The theories of ancient Greece **« 104–05** **dominated medical thinking for 2,000 years.**

GREEK MEDICAL THEORY

In ancient Greece (*c.* 500 BCE), it was believed that health depended on the body **maintaining a balance** of four fluids, or **"humors"**—blood, yellow bile, black bile, and phlegm. Each humor had its own characteristic qualities, combinations of hot, cold, wet, and dry. Blood, for example, was hot and wet. Disease was thought to arise when there was an imbalance of humors.

SKETCH FROM VESALIUS'S WORK

TEMPERAMENT

An **excess** of a particular humor was thought to give people their character, or **"temperament."** A sanguine temperament was believed to produce an excess of blood, which made the person cheerful, even-tempered, and optimistic, though feverish.

THE TRAVELS OF GREEK MEDICINE

These ideas spread from ancient Greece across the globe and dominated medicine for 2,000 years. They traveled across Europe during the Roman Empire, then spread to the Middle East, and to European colonies. While the core theories remained constant, doctors added to the Greek understanding of anatomy. In the 16th century, the Belgian physician **Andreas Vesalius**'s groundbreaking *On the Workings of the Human Body* contained detailed anatomical sketches.

In the 18th century, doctors researching human anatomy realized that some diseases, such as cancer, produced swellings in the solid parts of the body or changes to the appearance of organs such as the lungs and liver. The idea that disease was located in the solid parts of the body, not the fluids (see BEFORE), was firmly established by Parisian research into pathology in the early 19th century. In the city's huge hospitals, doctors carefully recorded the progress of many of the diseases suffered by hundreds of patients they treated. Those who died were dissected, to discover how their illness had affected the organs in their body. Using this approach, doctors were able to monitor the progress of many diseases, and to make precise distinctions between diseases of the heart and of the lungs, for example.

During the rest of the century, doctors explored the structure of the human body in finer detail. They described the various tissues that made up the organs and, using improved microscopes, began to explore the cells that formed the tissues. While some researchers studied the minute structures of the human body, others began to research the functions of the organs. By experimenting on animals, their patients, and sometimes themselves, doctors began to understand

19th-century surgeon's kit
Although antisepsis is often associated with the British surgeon Joseph Lister and the use of carbolic acid, safe surgery owed much to simple ideas—like making surgical instruments from steel, and sterilizing them after every operation.

Germ Warfare

In the 19th century, a number of significant developments in medical science greatly increased doctors' knowledge of anatomy and disease. Coupled with the transformation of hospitals into specialized treatment centers came a crucial understanding of the spread of infection and how to halt it.